Byunghyun (Ben) Ko

(703) 732-4572 • https://www.linkedin.com/in/bk1234/ • bko2@nd.edu

RESEARCH INTERESTS

- Computer vision and deep learning
- Visual representation learning and self-supervised methods
- Domain adaptation techniques
- Efficient and generalizable medical image analysis

I am particularly interested in developing robust models for understanding complex visual data, with applications in medical imaging, visual recognition, and multi-modal learning.

EDUCATION

University of Notre Dame, Notre Dame, IN — Expected May 2031
Ph.D. in Computer Science and Engineering

Northeastern University, Boston, MA — May 2026
M.S. in Computer Science — **CGPA**: 3.917/4.0

University of Notre Dame, Notre Dame, IN — May 2023
B.A. in Economics, Applied & Computational Mathematics & Statistics (ACMS) — **CGPA**: 3.832/4.0

PUBLICATIONS

Ko, B., Anisimov, A., Ke, K., Bharthepude, S., & Lee, J. "XSSR: Cross-Domain Self-Supervised Representative Selection for Efficient Annotation in Medical Image Segmentation" Accepted to the *International Conference on AI in Healthcare (AIiH)*, 2026. [arXiv]

Tian, A., **Ko, B.**, Qu, K., Liu, M., & Lee, J. "KLO-Net: A Dynamic K-NN Attention U-Net with CSP Encoder for Efficient Prostate Gland Segmentation from MRI" *SPIE Medical Imaging 2026: Image Processing*, 2026. [arXiv]

Ko, B., Tian, A., & Lee, J. "XAG-Net: A Cross-Slice Attention and Skip Gating Network for 2.5D Femur MRI Segmentation" *International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA)*, 2025. [arXiv] [code]

HONORS & AWARDS

Northeastern University – SP^2ARK PhD Pathway Apprenticeship ($4,000)	2026
Northeastern University – Khoury Research Apprenticeship ($7,500)	2025
Northeastern University – First Place, Student Research and Innovation Showcase	2025
KSEA UKC 2025 – Best Poster Award	2025
University of Notre Dame, College of Arts and Letters – Dean's List (6x)	2019 – 2023

RESEARCH EXPERIENCE

MIG Lab, Northeastern University — Jan 2025 – Apr 2026
Graduate Research Assistant – Advisor: Prof. Jeongkyu Lee

- Led development of XAG-Net, a 2.5D U-Net with novel pixel-wise cross-slice attention and skip attention gating blocks, attaining a Dice score of 0.9535 on femur segmentation while using 56% fewer parameters than 3D U-Net

- Conducted ablation studies analyzing the core components of XAG-Net, highlighting the primary role of cross-slice attention and its synergy with attention gating blocks
- Contributed to KLO-Net, a lightweight prostate segmentation model with dynamic K-NN attention and cross-stage partial blocks, achieving accuracy comparable to state-of-the-art models while reducing parameter count by 18x
- Developed XSSR, an annotation-efficient cross-domain medical image segmentation framework that uses self-supervised representation learning to reduce target-domain labeling requirements by 95% while maintaining competitive performance.

Department of Political Science, University of Notre Dame Sep 2019 – Mar 2020

Undergraduate Research Assistant – Advisor: Prof. Jeff Harden

- Collected and cleaned state-level policy data from 2013 to 2019 from *The Book of States* using Excel, contributing to an NSF-funded project on state data accessibility
- Researched and resolved statistical inconsistencies, documenting findings in structured text files to support analysis by the research team

TEACHING EXPERIENCE

Northeastern University May 2024 – Present

Graduate Teaching Assistant, Khoury College of Computer Sciences

- Fall 2025 – Database Management Systems (CS 5200)
- Summer 2025 – Pattern Recognition and Computer Vision (CS 5330)
- Spring 2025 – Data Science Capstone (DS 5500)
- Fall 2025 - Pattern Recognition and Computer Vision (CS 5330)
- Summer 2024 – Data Structures and Algorithms (CS 5008)

DEVELOPMENT EXPERIENCE

Exercise Form Analysis System [GitHub] Jan 2025 – May 2025

- Developed a web-based squat form analysis tool using RTMPose for efficient 2D pose estimation on uploaded videos, with user-selectable performance mode and camera views
- Designed and implemented view-specific metrics (knee angle and hip displacement) to assess squat depth and provide targeted feedback
- Built an interactive UI with Streamlit that outputs annotated GIFs, per-frame metrics, and summary feedback based on pose data

Pocket Caddy App [GitHub] Jan 2024 – May 2024

- Built an Android app in Java to recommend golf clubs based on distance, weather, and user attributes, using the MVVM architecture for consistent data handling
- Integrated chat functionality to the app with OpenAI API by utilizing Volley for network requests to send and receive messages

LEADERSHIP & OUTREACH

Vice President, Multimedia Information Group (MIG) Lab, Northeastern University 2025
Project Leader, Student International Business Council, University of Notre Dame 2021
Diversity Commissioner, Keenan Hall Council, University of Notre Dame 2019 – 2020